



**MASTERMIND
INDIA**

Spirit of India Special

Madhav SAWANT, 47

INNOVATION

**Light-emitting,
energy-saving tiles**

COST

Rs 600 per sq ft

WHAT IT DOES

**Saves electricity
by producing
light from glow-in-
the-dark paint**

By Nishika Patel

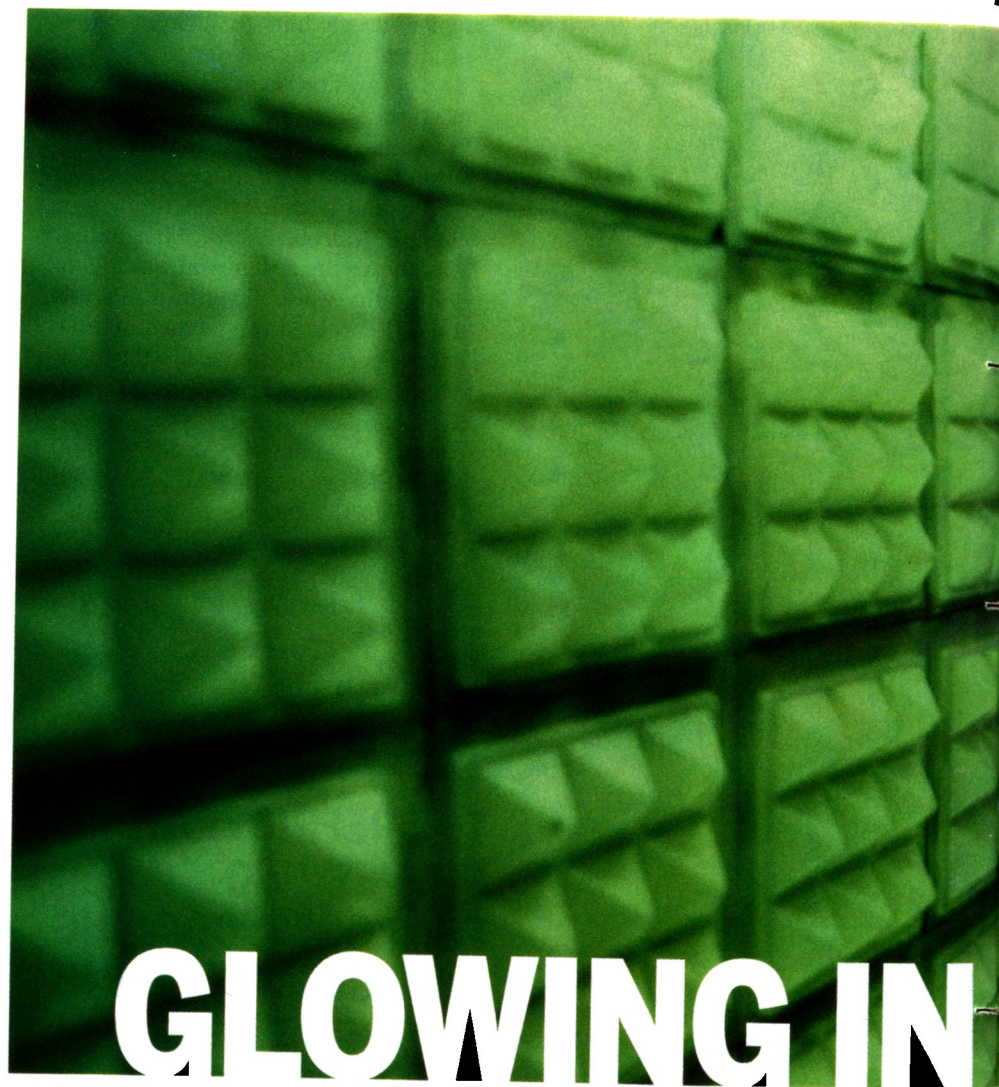
Paint manufacturer Madhav Sawant's glow-in-the-dark paint is used to light up the emergency exits of various buildings, but the mechanical engineer has a brighter, more lasting vision. The 47-year-old has developed a unique way to increase the brightness of glow powder so that it can be used as an alternative to lights and save energy on a large scale. With the help of IIT-Bombay, he has spent the last two years developing a four-inch wide transparent plastic tile, coated with glow powder. It features nine pyramids that increase the surface area and therefore the brightness. It emits green light

on its own for eight hours after absorbing artificial light for three hours. "The powder absorbs the light and stores it while the tile design increases the brightness," says Sawant. Over an area of one square metre, his glow-in-the-dark paint emits the equivalent of 2.7 candles while the tile emits the glow of 4.5 candles. Over an area of 100 sq ft, 900 tiles will produce 4W of light. Sawant says these tiles, which can be used on roads, highways, bridges and buildings, save electricity on a massive scale. "Up to 70 per cent of the electricity we use is to generate light. Even if we can salvage 5 per cent of this, it can be a big saving," he says.

The biggest challenge was getting the correct coating method on the tile. Sawant had to ensure that the coating,

which is made up of the glow pigment and the liquid, was in the right balance. "The pigment imparts the colour while the liquid helps it spread evenly. For good paint the pigment must be fine. But for night glow paint if the pigment is too fine it loses its property of emitting light," says the innovator. Sawant had to look at unconventional methods of coating and racked his brains, testing 14 methods before he came up with the perfect concoction. He designed a special machine to spray the glow paint. Sawant has just completed the research with the help of a Rs 5-lakh grant from the government and is looking for investors to commence production.

Sawant's love for innovation goes back to this college days. After



GLOWING IN